GRAPHS AND DISCUSSION

1. CORRELATION MATRIX OF PHYSIOLOGICAL VARIABLES

The correlation matrix (Fig.1) provides a detailed visualization of the relationships between all numerical variables used in the dataset. It illustrates the strength and direction of correlation between parameters such as age, activity levels, and muscle strength. Positive correlation indicates that both variables increase together, while negative correlation indicates an inverse relationship. This analysis helps in identifying the most influential features affecting muscle performance and fatigue. Highly correlated variables can also be useful for model development, while weak correlations indicate lesser impact on the output. The heatmap representation enhances interpretability by using color gradients to distinguish correlation strength, thereby supporting effective feature analysis.

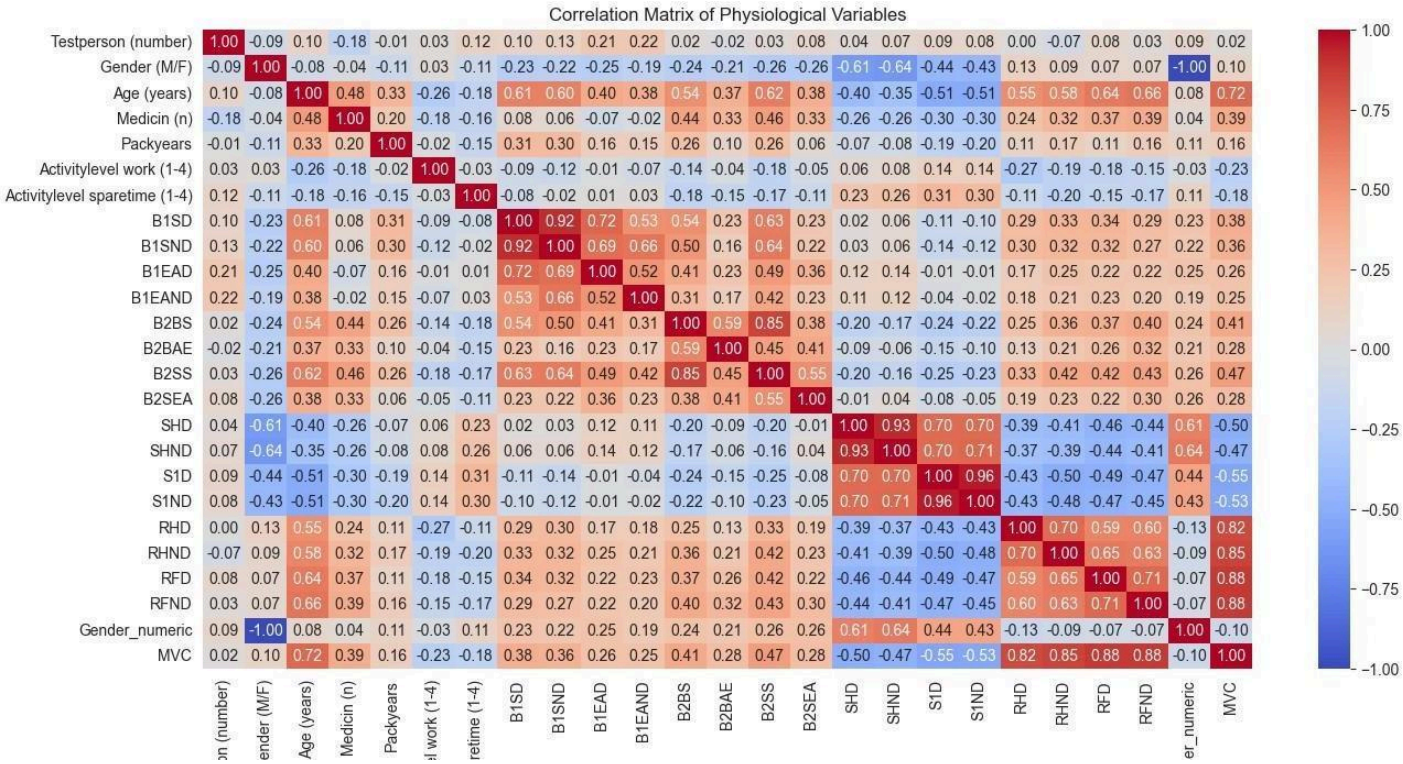

Fig.1 Correlation matrix

2. DISTRIBUTION OF MAXIMUM VOLUNTARY CONTACTION

The distribution graph (Fig.2) shows that most of the MVC values are concentrated in the range of approximately 550 to 750, indicating that a large proportion of subjects have moderate muscle strength. The peak of the distribution occurs around 650 to 700, which

represents the most common MVC values in the dataset. The curve gradually decreases towards higher values, forming a right-skewed distribution, suggesting that fewer individuals have very high muscle strength (above 900). Similarly, a small number of subjects fall in the lower range (below 500), indicating weaker muscle strength. Overall, the graph demonstrates that while the majority of individuals exhibit average muscle performance, there is noticeable variability with some extreme high and low values present in the dataset.

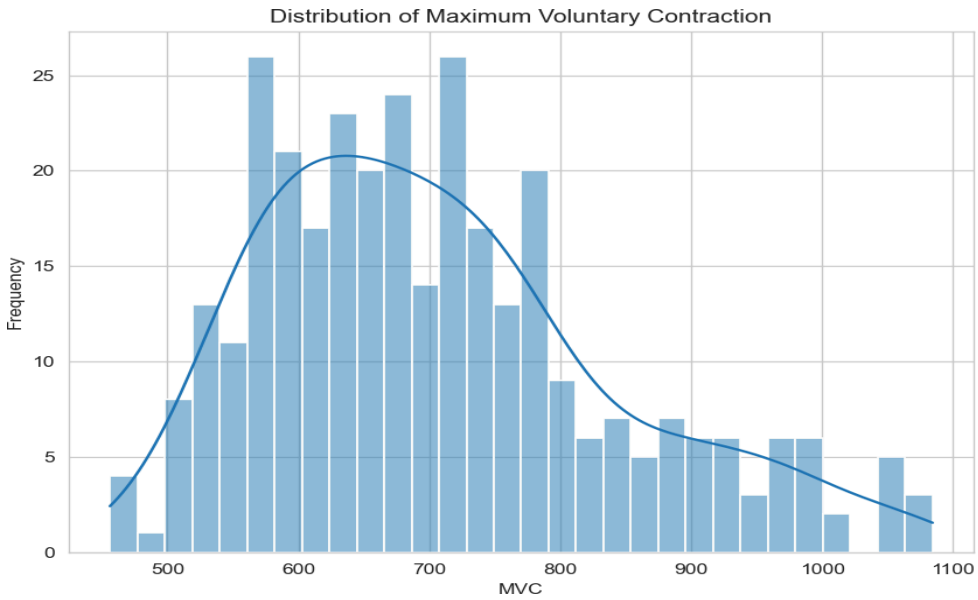

Fig.2 Distribution of MVC

3. SIGNAL CONDITIONING AND STABILITY

This graph (Fig.3) is a box plot with a swarm plot that shows how Maximum Voluntary Contraction values are distributed across three age groups: Young, Middle, and Older. The box plot represents the spread of data using median and quartiles, while the swarm plot displays individual data points for each subject. It allows comparison of muscle strength variations between different age groups in the dataset.

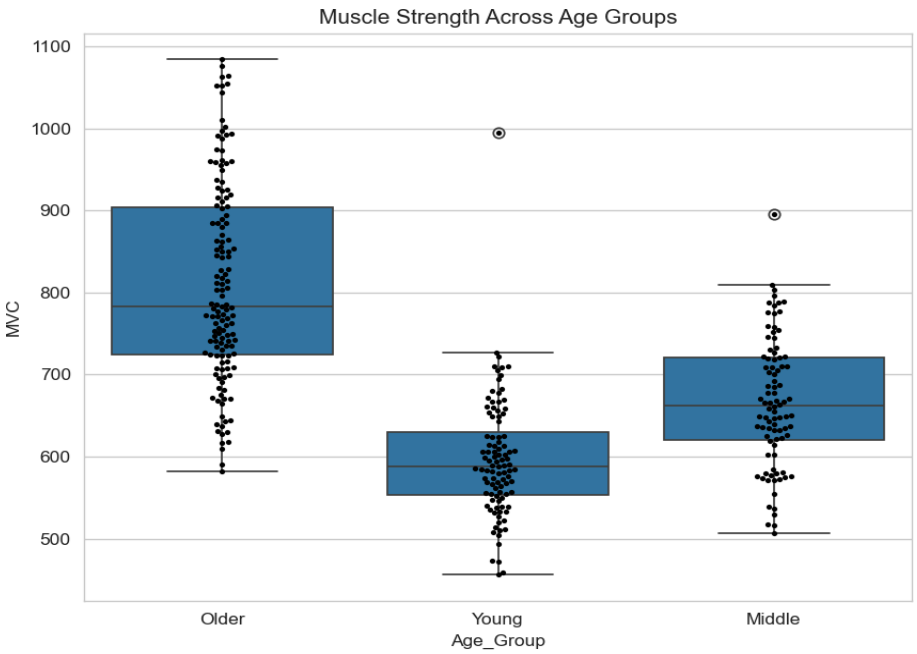

Fig.3 Muscle strength across age groups

4. MUSCLE STRENGTH BY GENDER

This graph (Fig.4) is a box plot with a swarm plot that shows the distribution of Maximum Voluntary Contraction values for different genders in the dataset. It compares muscle strength between male and female subjects by displaying the median, spread, and individual data points for each group. The visualization helps in identifying differences in muscle strength levels between genders and the variation within each group.

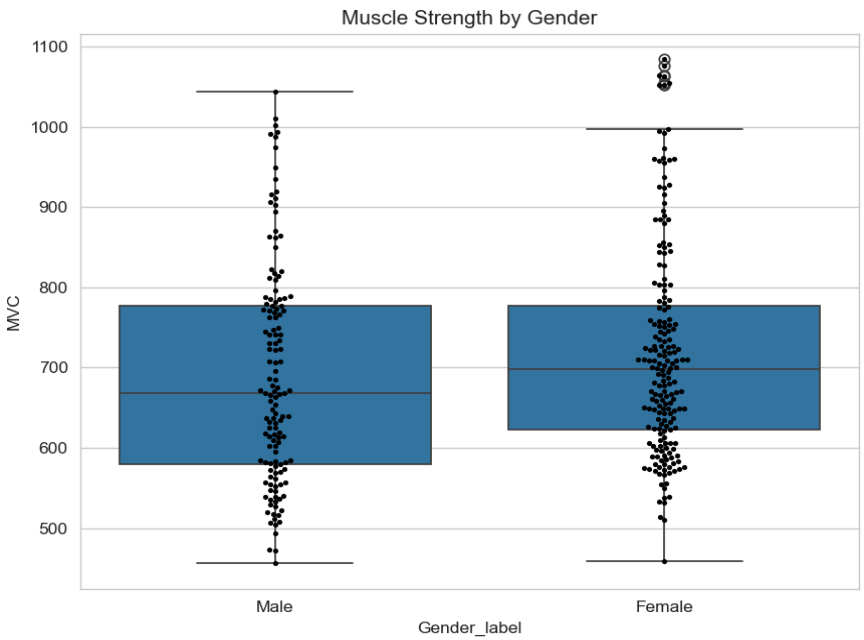

Fig.4 Muscle strength

5. EFFECT OF WORK ACTIVITY

From the graph (Fig.5), it can be inferred that muscle strength varies with work activity level, but not in a strictly increasing or decreasing pattern. Subjects with very low activity levels (level 0/1) show a relatively higher median MVC, indicating some individuals with good strength despite low physical work. At moderate activity levels (level 2 and 3), the MVC values are more widely spread, showing high variability among individuals.

At the highest activity level (level 4), MVC values appear moderately high but with fewer data points, indicating limited samples. Overall, the graph suggests that while activity level does influence muscle strength, other factors such as age and lifestyle also play a significant role, leading to variability within each group.

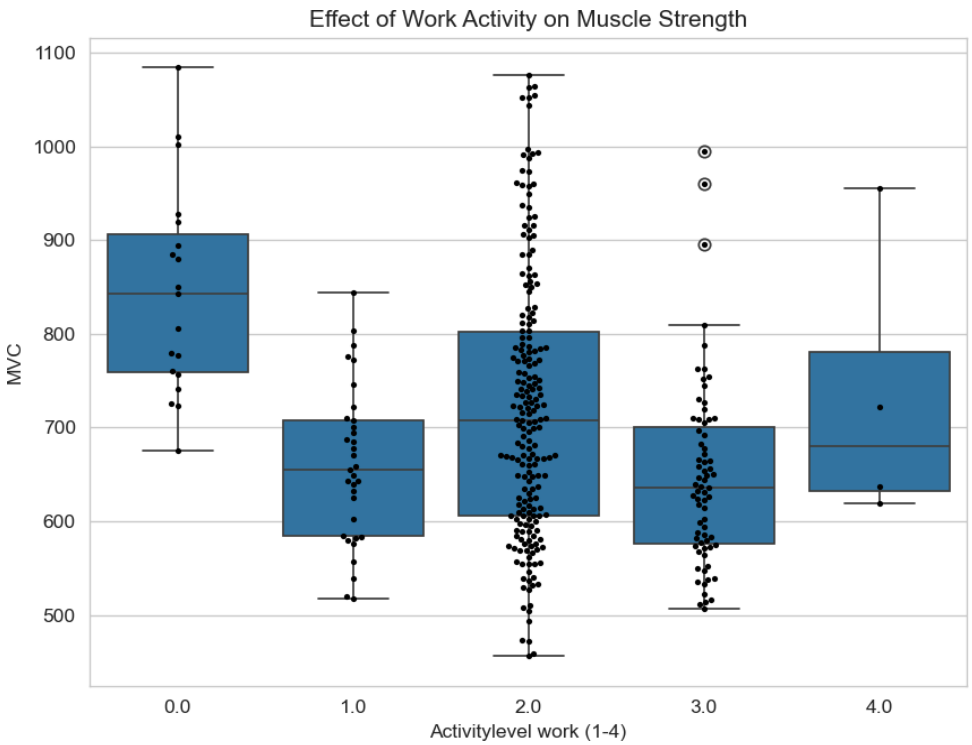

Fig.5 Effect of Work Activity

6. EFFECT OF SPARE TIME

This graph (Fig.6) shows the relationship between recreational (spare-time) activity levels (1–4) and muscle strength. From the graph, it can be inferred that individuals with lower to moderate recreational activity levels (levels 1–3) tend to have higher median MVC values, indicating better muscle strength. In particular, activity level 1 shows a relatively higher median and a wider spread,

suggesting that some individuals in this group possess very high strength values. As the recreational activity level increases to level 4, the median MVC appears to decrease, indicating comparatively lower muscle strength in this group.

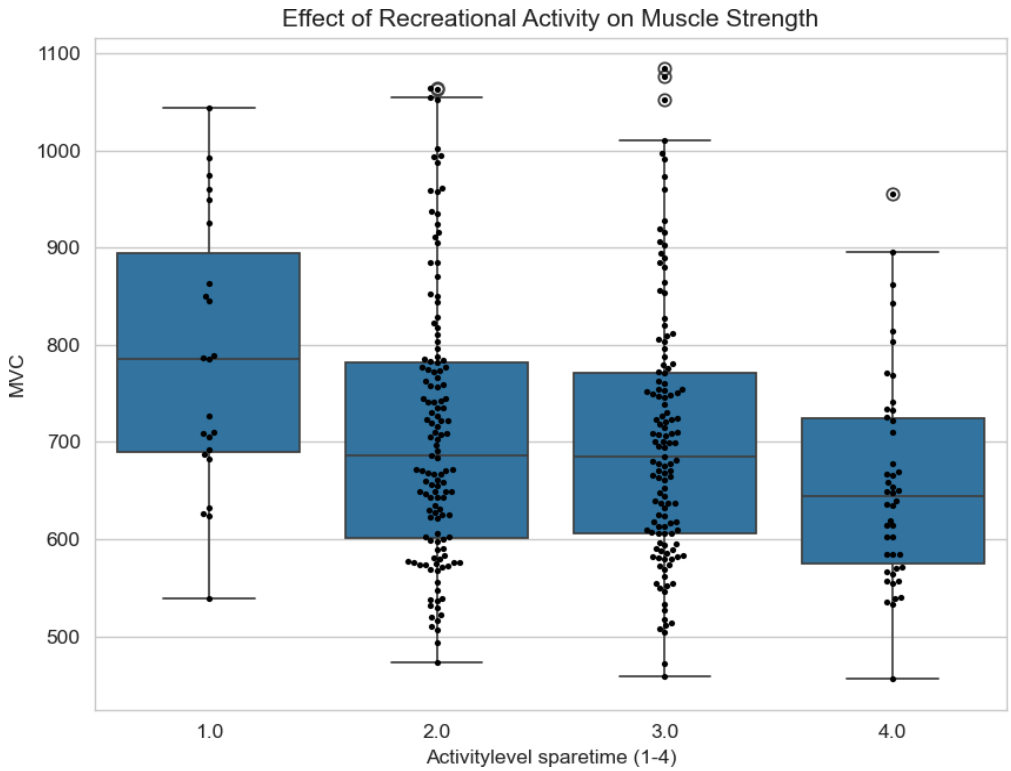

Fig.6 Effect of recreational Activity on muscle strength

7. FEATURE IMPORTANCE

The graph (Fig.7) illustrates the importance of different input features in predicting Maximum Voluntary Contraction within our model. It clearly shows that age is the most dominant factor, contributing the largest share to the prediction, which suggests that muscle strength (MVC) is highly influenced by aging related physiological changes. Packyears have a moderate impact, indicating that smoking history may negatively affect muscle performance. Activity levels, both during work and spare time, have smaller but still noticeable contributions, suggesting that physical activity plays a role, though less significant than age. Gender has the least influence among the features. Overall, the model appears to depends heavily on age, with lifestyle factors providing secondary adjustments, highlighting a strong physiological basis for MVC variation in the dataset.

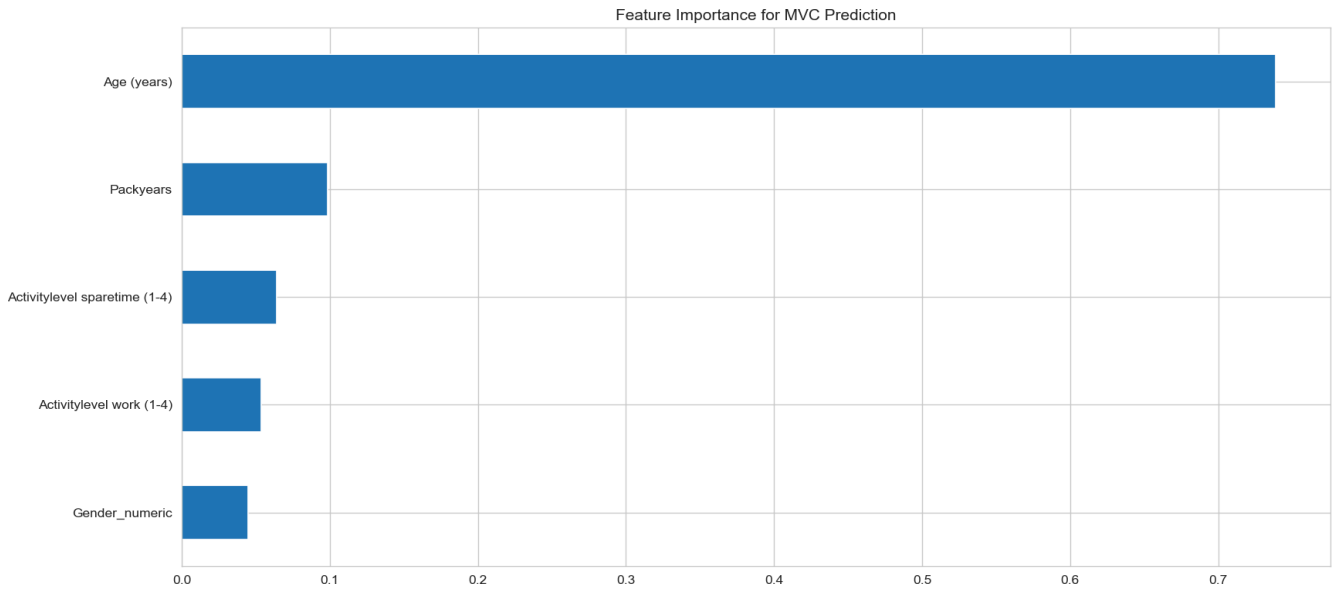

Fig.7 Importance of different features

8. STRENGTH vs ENDURANCE PLOT

Fig.8 is a scatter plot illustrating the relationship between muscle strength, represented by Maximum Voluntary Contraction, and endurance, represented by safe exercise time, across different age groups. Each data point corresponds to an individual, with colors distinguishing young, middle-aged, and older subjects. Individuals with higher MVC values tend to have lower safe exercise times, indicating that greater muscle strength does not necessarily mean higher endurance. Additionally, younger individuals tend to have higher endurance levels, while older individuals show reduced endurance. Middle-aged individuals fall between these two groups, demonstrating a shifting pattern. Overall, the graph highlights how both age and muscle strength influence endurance, supporting the need for personalized exercise recommendations.

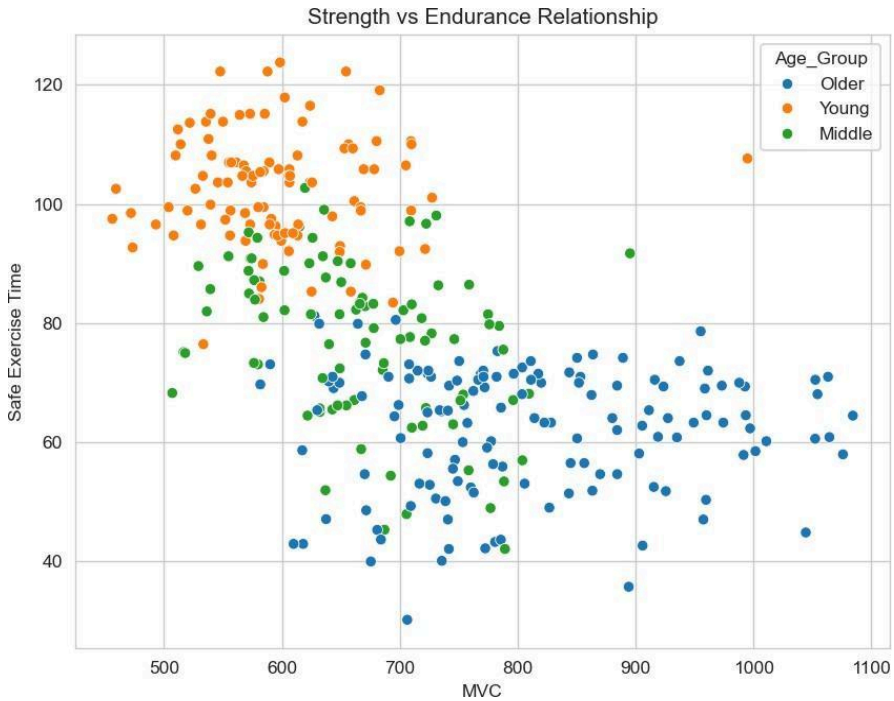

Fig.8 Scatter plot of Strength vs Endurance